

Computer Architecture Lab 10 Report

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实验 10 wujian 平台介绍和 FPGA 调试

1. 实验目的

了解 wujian100 SoC 的组成架构, 掌握 wujian 平台的前端仿真和 FPGA 调试.

2. 实验步骤

2.1. 依照“二、wujian100 SoC 的前端仿真”的步骤完成 case/ 目录下的示例的 timer_test.c 仿真, 对测试结果进行截图.

1. 实验环境

连接实验服务器, 获取实验相关文件并配置实验环境.

```
chenyiheng@ubuntuv80-CS5280H2: ~/g13g14/z24301050026/lab10/wujian100_open/tools
jz2024@JZ2024:~$ ssh ComputerArchitectureLab10
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

301 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Thu Jun 11 15:57:40 2026 from 10.219.252.115
chenyiheng@ubuntuv80-CS5280H2:~$ cd g13g14/
chenyiheng@ubuntuv80-CS5280H2:~/g13g14$ mkdir z24301050026
chenyiheng@ubuntuv80-CS5280H2:~/g13g14$ cd z24301050026/
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026$ cp -r /home/data/sunxuejian/XXepic/Soc2026/lab10 ./
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026$ ll lab10/
total 24
drwxr-xr-x  6 chenyiheng nas 4096 6月 11 16:23 ./
drwxr-xr-x  3 chenyiheng nas 4096 6月 11 16:23 ../
drwxr-xr-x  2 chenyiheng nas 4096 6月 11 16:23 elf2mem_tool/
drwxr-xr-x  9 chenyiheng nas 4096 6月 11 16:23 riscv_toolchain/
drwxr-xr-x 13 chenyiheng nas 4096 6月 11 16:23 wujian100_open/
drwxr-xr-x  5 chenyiheng nas 4096 6月 11 16:23 wujian100_p2e/
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026$ cd lab10/wujian100_open/
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open$ ls
case doc fpga lib LICENSE README.md regress sdk soc tb tools workdir
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open$ cd tools/
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/tools$ ls
run_case setup.csh setup.sh Srec2vmem Srec2vmem.py
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/tools$ source setup.sh
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/tools$ echo $VIVADO_HOME
/home/data/xilinx/Vivado/2024.2
chenyiheng@ubuntuv80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/tools$
```

图 1: Experiment Environment

2. 前端仿真

进入实验目录, 运行 GalaxSim 仿真, 进行 timer_test.c 的仿真测试.

```

chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/workdir$ ls
case doc fpga lib LICENSE README.md regress sdk soc tb tools workdir
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open$ cd workdir/
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/workdir$ ls
config.h  fprintf.c  getc.o  _lltostr.o  printf.o  rtl_gpr.log  timer_test.c  __v_printf.c  vsprintf.o
crt0.o    fprintf.o  _isinf.c  _lltostr.c  putc.c    run_case.report  timer_test.elf  vprintf.c  vtimer.h
crt0.s    fputc.c   _isinf.o  _lltostr.o  putchar.c  snprintf.c       timer_test.hex  __v_printf.o  xsim
datatype.h fputc.o   _isnan.c  Makefile    putchar.o  snprintf.o       timer_test.o    vprintf.o    xsim.data
designInfo getc.c    _isnan.o  minilibc_stdio.h  putc.o    sprintf.c        timer_test.obj  vsnprintf.c  xsim.src
__dtostr.c getchar.c linker.lcf printf.c    puts.c    sprintf.o        vfprintf.c     vsnprintf.o   xsim.usr_libs
__dtostr.o getchar.o _lltostr.c printf.h    puts.o    test.pat         vfprintf.o     vsprintf.c
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/workdir$ ./tools/run_case -sim_tool galaxsim ../case/timer/timer_test.c
Useless use of not in void context at ../tools/run_case line 207.

Step1 (Remove all things in current 'workdir') is finished!
../case/timer/timer_test.c

Step2 (Process the command line arguments) is finished!
mkdir: cannot create directory './regress/regress_result': File exists
rm -rf *.o *.pat *.elf *.obj *.hex
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o crt0.o crt0.s
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o __dtostr.o __dtostr.c
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o getchar.o getchar.c
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o printf.o printf.c
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o vprintf.o vprintf.c
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o sprintf.o sprintf.c
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o snprintf.o snprintf.c
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o vfprintf.o vfprintf.c
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o fprintf.o fprintf.c
../riscv_toolchain/bin/riscv64-unknown-elf-gcc -c -march=rv32emc_xthead -mabi=ilp32e -De902 -Wa,--defsym=e902=1 -O3 -funroll
-all-loops -fgcse-sm -finline-limit=500 -fno-schedule-insns -ffunction-sections -fdata-sections -o timer_test.o timer_test.c
In file included from timer_test.c:12:

```

图 2: GalaxSim Simulation

观察仿真结果, 可以看到程序正确输出了 Hello! 字样, 说明前端仿真成功.

```

chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/workdir$
Compiling module chregc4
Compiling module chregc5
Compiling module chregc6
Compiling module chregc7
Compiling module chregc8
Compiling module chregc9
Compiling module chregc10
Compiling module chregc11
Compiling module chregc12
Compiling module chregc13
Compiling module chregc14
Compiling module chregc15
Compiling module wdt_biu
Compiling module cr_elic_reg_if
Compiling module sms_sram_bk2
Compiling module timers_frc
Compiling module fpga_spram
Compiling module fpga_byte_spram
Total 194 modules compiled.
Fatal(0), Error(0).
Time: 7.69s Mem: 222572k to compile
Time: 0.53s Mem: 114576k to elab
Time: 0.32s Mem: 44268k to link
==> Exec cmd: /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_open/workdir/xsim -R
GalaxSim (R) v24.09.00.a055 x86_64_R Dec.25 2024
#####time: 0, Dump start#####
output to test.xedb
*****START TO LOAD PROGRAM*****

Hello!

timer t*****simulation finish for meet max simulation time*****

$finish called from file "../tb/tb.v", line 314.
$finish at simulation time 150000000
Simulation cycle time: 15000000000 ps
Simulation CPU time: 28.1 s
Finish time: 四 6月 11 16:29:04 2026

Step4 (Run simulation) is finished
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_open/workdir$

```

图 3: Simulation Results

2.2. 依照“三、wujian100 SoC 的 FPGA 仿真测试”的步骤完成示例的项目测试, 对测试结果进行截图 (请截取不同于示例部分的波形图片), 并说说捕获的波形是如何得到的, 为什么用这种方法来捕捉.

1. 项目编译

在 Windows 系统下安装实验所需的 CDK 软件. 修改示例项目中的 `putc()` 实现后, 编译成 `.elf` 文件.

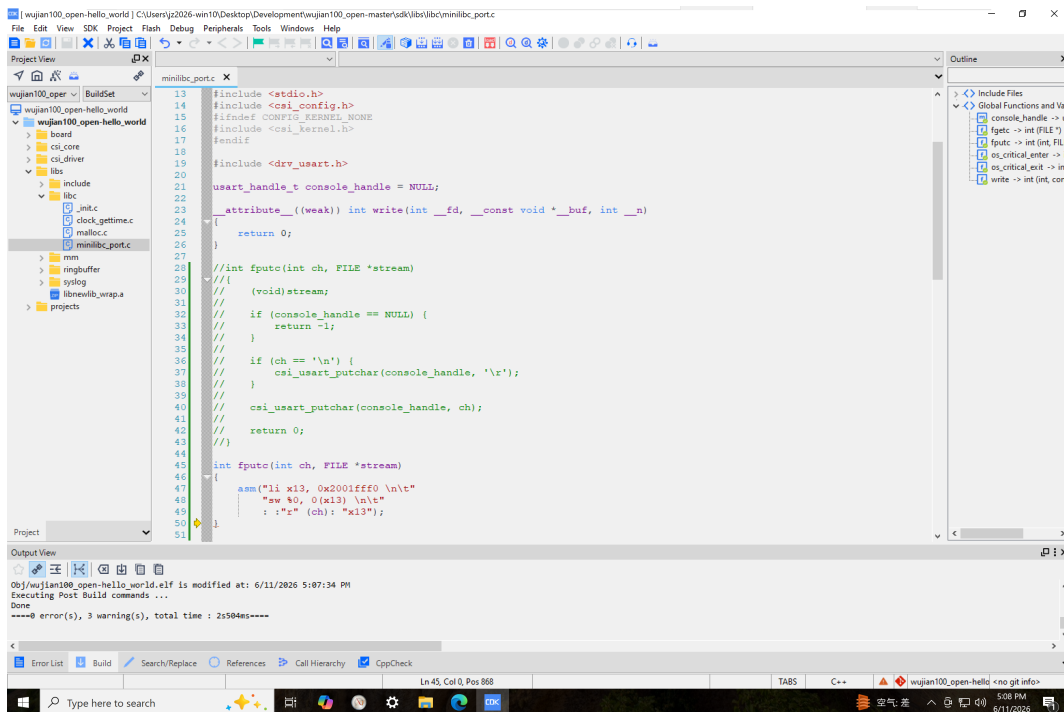


图 4: Compile ELF

2. 文件转换

将编译好的 `.elf` 文件上传到服务器上, 并使用 `elf2mem_tool/` 目录下的 Python 脚本进行转换和分割.



图 5: Convert ELF to MEM

3. vSyn 综合

进入 wujian100_p2e/ 目录进行 FPGA 测试. 将分割后的 .mem 文件复制到 src/ 目录下, 并执行 ./run_tb.sh 进行 vSyn 综合.

```
chenyiheng@ubuntu20-08:~/v24.09.01.s003/x86_64/Dec.20.2024$ ls
etc hw-e1.hdf PNR_settings.tcl run_tb.sh src vcom_compile.tcl vdbg.tcl
chenyiheng@ubuntu20-08:~/v24.09.01.s003/x86_64/Dec.20.2024$ cp ../elf2mem_tool/hello_world_b* ./src/
chenyiheng@ubuntu20-08:~/v24.09.01.s003/x86_64/Dec.20.2024$ ls src/
hello_world_b0.mem hello_world_b2.mem params tb wujian100_open-master
hello_world_b1.mem hello_world_b3.mem sim_lib wujian100_open_lib_for_p2e.lst wujian100_open_syn_for_p2e.lst
chenyiheng@ubuntu20-08:~/v24.09.01.s003/x86_64/Dec.20.2024$ ./run_tb.sh
VSYN
v24.09.01.s003 x86_64 Dec.20 2024
Copyright © 2020-2024 by XEPIC Co., Ltd.
ALL RIGHTS RESERVED https://www.x-epic.com
This program is proprietary and confidential information of XEPIC Co., Ltd.
and may be used and disclosed only as authorized in a license agreement
controlling such use and disclosure.
VSYN v24.09.01.s003 x86_64 Dec.20 2024
Start at Thu Jun 11 17:29:28 2026
Command: /opt/xepic/hpe-24.09.01.s003/bin/vsyn ./src/tb/stb.v -f ./src/wujian100_open_lib_for_p2e.lst -f ./src/wujian100_open_syn_for_p2e.lst -o ./src/x_wujian100_open_tb.v
Using HPE_HOME "/opt/xepic/hpe-24.09.01.s003"
-- Parsing command file ./src/wujian100_open_lib_for_p2e.lst
-- Parsing command file ./src/wujian100_open_syn_for_p2e.lst
-- Analyzing Verilog file ./src/tb/stb.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/sim_lib/PAD_DIG_IO.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/sim_lib/PAD_OSC_IO.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/sim_lib/fpga_byte_spram.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/sim_lib/fpga_spram.v
-- Analyzing Verilog file ./src/sim_lib/STD_CELL.v
-- Analyzing Verilog file ./src/wujian100_open-master/fpga/wujian100_open_fpga_top.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/ahb_matrix_top.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/smu_top.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/sms.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/ls_sub_top.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/retu_top.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/tim5.v
-- Analyzing Verilog file ./src/wujian100_open-master/soc/tim.v
INFO VSYN-11328 ./src/wujian100_open-master/soc/tim.v:94: analyzing included file ./src/params/timers_params.v
INFO VSYN-12320 ./src/wujian100_open-master/soc/tim.v:94: back to file ./src/wujian100_open-master/soc/tim.v
INFO VSYN-11328 ./src/wujian100_open-master/soc/tim.v:537: analyzing included file ./src/params/timers_params.v
INFO VSYN-12320 ./src/wujian100_open-master/soc/tim.v:537: back to file ./src/wujian100_open-master/soc/tim.v
```

图 6: vSyn Synthesis

完成后打开 vsyn.log, 确认无错误.

```
chenyiheng@ubuntu20-08:~/v24.09.01.s003/x86_64/Dec.20.2024$ cat vsyn.log
962 -- ./src/wujian100_open-master/soc/sim_lib/PAD_DIG_IO.v:11: finish compiling module 'PAD_DIG_IO'
963 -- ./src/wujian100_open-master/fpga/wujian100_open_fpga_top.v:39: [PERF] Synthesis module work.wujian100_open_top start at 2026-06-11 17:31:22, CPU time: 115.59s, Wall time: 114s, Max memory usage: 346.11MB
964 -- ./src/wujian100_open-master/fpga/wujian100_open_fpga_top.v:39: [PERF] Synthesis module work.wujian100_open_top finish at 2026-06-11 17:31:22, CPU time: 115.60s (+ 0.01 s), Wall time: 114s (+ 0 s), Max memory usage: 346.11MB
965 -- ./src/wujian100_open-master/fpga/wujian100_open_fpga_top.v:39: finish compiling module 'wujian100_open_top'
966 -- ./src/tb/stb.v:45: defer synthesizing module 'wujian100_open_stb'
967 -- ./src/tb/stb.v:45: finish compiling module 'wujian100_open_stb'
968 -- ./src/sim_lib/STD_CELL.v:11: compiling module 'Standard_Cell_CLK_GATE'
969 WARN VSYN-12580 ./src/sim_lib/STD_CELL.v:26: latch inferred for net 'q_tmp'
970 -- ./src/sim_lib/STD_CELL.v:11: [PERF] Synthesis module work.Standard_Cell_CLK_GATE start at 2026-06-11 17:31:22, CPU time: 115.60s, Wall time: 114s, Max memory usage: 346.11MB
971 -- ./src/sim_lib/STD_CELL.v:11: [PERF] Synthesis module work.Standard_Cell_CLK_GATE finish at 2026-06-11 17:31:22, CPU time: 115.61s (+ 0.01 s), Wall time: 114s (+ 0 s), Max memory usage: 346.11MB
972 -- ./src/sim_lib/STD_CELL.v:11: finish compiling module 'Standard_Cell_CLK_GATE'
973 -- ./src/sim_lib/STD_CELL.v:38: compiling module 'Standard_Cell_CLK_MUX2'
974 -- ./src/sim_lib/STD_CELL.v:38: [PERF] Synthesis module work.Standard_Cell_CLK_MUX2 start at 2026-06-11 17:31:22, CPU time: 115.61s, Wall time: 114s, Max memory usage: 346.11MB
975 -- ./src/sim_lib/STD_CELL.v:38: [PERF] Synthesis module work.Standard_Cell_CLK_MUX2 finish at 2026-06-11 17:31:22, CPU time: 115.62s (+ 0.01 s), Wall time: 114s (+ 0 s), Max memory usage: 346.11MB
976 -- ./src/sim_lib/STD_CELL.v:38: finish compiling module 'Standard_Cell_CLK_MUX2'
977 -- ./src/wujian100_open-master/soc/core_top.v:11: [PERF] Synthesis module work.core_top start at 2026-06-11 17:31:22, CPU time: 115.63s, Wall time: 114s, Max memory usage: 346.11MB
978 -- ./src/wujian100_open-master/soc/core_top.v:11: [PERF] Synthesis module work.core_top finish at 2026-06-11 17:31:22, CPU time: 115.65s (+ 0.02 s), Wall time: 114s (+ 0 s), Max memory usage: 346.11MB
979 -- ./src/tb/stb.v:45: [PERF] Synthesis module work.wujian100_open_stb start at 2026-06-11 17:31:22, CPU time: 115.65s, Wall time: 114s, Max memory usage: 346.11MB
980 -- ./src/tb/stb.v:45: [PERF] Synthesis module work.wujian100_open_stb finish at 2026-06-11 17:31:22, CPU time: 115.65s (+ 0.00 s), Wall time: 114s (+ 0 s), Max memory usage: 346.11MB
981 -- [PERF] Elaborate finish at 2026-06-11 17:31:22, CPU time: 115.66s (+ 114.54 s), Wall time: 114s (+ 113 s), Max memory usage: 346.11MB
982 -- Writing Verilog netlist file './src/x_wujian100_open_tb.v'
983 Total modules: 193; Total instances: 545;
984 Total LUT: 34255; Total DSP: 0; Total RAM: 22; Total CARRY: 323; Total Register: 15358; Total gate count: 253208;
985 -- Writing area report to file 'vsyn_area.rpt'
986 -- [PERF] Visit netlist finish at 2026-06-11 17:31:22, CPU time: 116.19s (+ 0.53 s), Wall time: 114s (+ 0 s), Max memory usage: 346.11MB
987 -- [PERF] Output scripts finish at 2026-06-11 17:31:22, CPU time: 116.19s (+ 0.00 s), Wall time: 114s (+ 0 s), Max memory usage: 346.11MB
988 CPU time: 116.190000s, Wall time: 114.491218s, Max memory usage: 346.11MB.
~
:set nu 988,1 Bot
```

图 7: Synthesis Results

4. vCom 编译

修改 vdbg.tcl 中定义的 .mem 文件路径, 实验文档中将需要修改的文件名称误写成了 tb/stb.sv. 随后执行 vcom vcom_compile.tcl 进行 vCom 编译.

```
chenyiheng@ubuntu80-CS5280H2: ~/g13g14/z24301050026/lab10/wujian100_p2e
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ ls
etc hw-e1.hdf PNR_settings.tcl run_tb.sh src vcom_compile.tcl vdbg.tcl vsyn_area.rpt vsyn.log
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ vim vsyn.log
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ ls src/tb/
1_stb.v busmnt.v stb.sv tb.v virtual_counter.v
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ vim src/tb/stb.sv
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ grep -rn "memory -write" .
./vdbg.tcl:10:memory -write -format %readmemh -name wujian100_open_stb.x_wujian100_open_top.x_retu_top.x_smu_top.x_sms_top.x_isr
am_top.x_sms_sram.x_fpga_sram.x_fpga_byte3_sram.mem -file /home/data/sunxuejian/XXepic/Soc2026/lab10/wujian100_p2e_new/src/he
lo_world_b3.mem -start 0x0
./vdbg.tcl:11:memory -write -format %readmemh -name wujian100_open_stb.x_wujian100_open_top.x_retu_top.x_smu_top.x_sms_top.x_isr
am_top.x_sms_sram.x_fpga_sram.x_fpga_byte2_sram.mem -file /home/data/sunxuejian/XXepic/Soc2026/lab10/wujian100_p2e_new/src/he
lo_world_b2.mem -start 0x0
./vdbg.tcl:12:memory -write -format %readmemh -name wujian100_open_stb.x_wujian100_open_top.x_retu_top.x_smu_top.x_sms_top.x_isr
am_top.x_sms_sram.x_fpga_sram.x_fpga_byte1_sram.mem -file /home/data/sunxuejian/XXepic/Soc2026/lab10/wujian100_p2e_new/src/he
lo_world_b1.mem -start 0x0
./vdbg.tcl:13:memory -write -format %readmemh -name wujian100_open_stb.x_wujian100_open_top.x_retu_top.x_smu_top.x_sms_top.x_isr
am_top.x_sms_sram.x_fpga_sram.x_fpga_byte0_sram.mem -file /home/data/sunxuejian/XXepic/Soc2026/lab10/wujian100_p2e_new/src/he
lo_world_b0.mem -start 0x0
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ vim vdbg.tcl
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ vim vdbg.tcl
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ vim vdbg.tcl
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ grep -rn "memory -write" .
./vdbg.tcl:10:memory -write -format %readmemh -name wujian100_open_stb.x_wujian100_open_top.x_retu_top.x_smu_top.x_sms_top.x_isr
am_top.x_sms_sram.x_fpga_sram.x_fpga_byte3_sram.mem -file /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/src/he
llo_world_b3.mem -start 0x0
./vdbg.tcl:11:memory -write -format %readmemh -name wujian100_open_stb.x_wujian100_open_top.x_retu_top.x_smu_top.x_sms_top.x_isr
am_top.x_sms_sram.x_fpga_sram.x_fpga_byte2_sram.mem -file /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/src/he
llo_world_b2.mem -start 0x0
./vdbg.tcl:12:memory -write -format %readmemh -name wujian100_open_stb.x_wujian100_open_top.x_retu_top.x_smu_top.x_sms_top.x_isr
am_top.x_sms_sram.x_fpga_sram.x_fpga_byte1_sram.mem -file /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/src/he
llo_world_b1.mem -start 0x0
./vdbg.tcl:13:memory -write -format %readmemh -name wujian100_open_stb.x_wujian100_open_top.x_retu_top.x_smu_top.x_sms_top.x_isr
am_top.x_sms_sram.x_fpga_sram.x_fpga_byte0_sram.mem -file /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/src/he
llo_world_b0.mem -start 0x0
chenyiheng@ubuntu80-CS5280H2:~/g13g14/z24301050026/lab10/wujian100_p2e$ vcom vcom_compile.tcl
Checkout license successfully. The remaining valid days: 6383.
VCOM
v24.09.01.s003 x86_64 Dec.20 2024

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ALL RIGHTS RESERVED https://www.x-epic.com
```

图 8: vCom Compilation

完成后打开 vcom.log, 确认无错误.

```
chenyiheng@ubuntu80-CS5280H2: ~/g13g14/z24301050026/lab10/wujian100_p2e
869 [2026-06-11 17:53:11] [3149297:info] 31000 Creating debug ip (sys_ctrl_top_phc)
870 [2026-06-11 17:53:11] [3149297:info] 30300 SA/0: transformation...
871 [2026-06-11 17:53:11] [3149297:info] 30300 SA/0: number of transform logics: 57
872 [2026-06-11 17:53:11] [3149297:info] 51700 Transformation finished, start to connect loader ports to new driver ports
873 [2026-06-11 17:53:11] [3149297:info] 30300 SA/0: succ_num 57, err_num 0
874 [2026-06-11 17:53:11] [3149297:info] 30500 NP/0: processing...
875 [2026-06-11 17:53:11] [3149297:info] 30500 NP/0: writing npdb 'np_0.db'
876 [2026-06-11 17:53:11] [3149297:info] 10000 [DG/Imp] SUMMARY: CPU time: 0.20 s, Wall time: 0.03 s, Max memory usage: 646.89
MB.
877 [2026-06-11 17:53:11] [3149297:info] 30110 Insert clock control unit ...
878 [2026-06-11 17:53:11] [3149297:info] 30110 Gclk index is 1, pin is CA36
879 [2026-06-11 17:53:11] [3149297:info] 30110 Add physical constraints ...
880 [2026-06-11 17:53:11] [3149297:info] 30222 Number of removed VVAC dummy ICGU instance is: 0
881 [2026-06-11 17:53:11] [3149297:info] 30222 Number of EXPORT related inserted instance is: 0
882 [2026-06-11 17:53:11] [3149297:info] 30222 Number of EXPORT data out clock-2-data FD instance is: 0
883 [2026-06-11 17:53:11] [3149297:info] 30110 DG post trasformation ...
884 [2026-06-11 17:53:11] [3149297:info] 30300 SA/0: processed no driver common net '2', clock net '0'
885 [2026-06-11 17:53:11] [3149297:info] 30300 SA/0: writing sadb 'sa_0.db'
886 [2026-06-11 17:53:11] [3149297:info] 30200 Debug ip connecting ...
887 [2026-06-11 17:53:11] [3149297:info] 30701 MA: partition: 0: number of access memory instance: 0
888 [2026-06-11 17:53:11] [3149297:info] 91919 Prepare Memory Access DB, partition ID: 0, XPM memory access total number: 4
889 [2026-06-11 17:53:11] [3149297:info] 91919 Prepare Memory read back DB, partition ID: 0, XPM memory total number: 22
890 [2026-06-11 17:53:11] [3149297:info] 10000 [DG/PostTran] SUMMARY: CPU time: 0.28 s, Wall time: 0.02 s, Max memory usage: 64
6.89 MB.
891 [2026-06-11 17:53:11] [3149297:info] 30110 Remove tag breaker ...
892 [2026-06-11 17:53:11] [3149297:info] 30601 dynamic trigger speed level 0
893 [2026-06-11 17:53:11] [3149297:info] 30699 Start to dump clock constraint file.
894 [2026-06-11 17:53:11] [3149297:info] 30000 DG: Process partition 0 end
895 [2026-06-11 17:53:11] [3149297:info] 10000 [DesignImplementation] SUMMARY: CPU time: 2.01 s, Wall time: 0.28 s, Max memory
usage: 669.19 MB.
896 [2026-06-11 17:53:12] [3149297:info] 30120 Write info files ...
897 [2026-06-11 17:53:12] [3149297:info] 34000 Remove external ports
898 [2026-06-11 17:53:12] [3149297:info] 30120 Check driverless ports ...
899 [2026-06-11 17:53:12] [3149297:info] 10000 [NetlistDumping] SUMMARY: CPU time: 1.02 s, Wall time: 0.84 s, Max memory usage:
675.11 MB.
900 [2026-06-11 17:53:12] [3149297:info] 30000 DG: Design Generation End
901 [2026-06-11 17:53:12] [3149297:info] 10000 [TOTAL] SUMMARY: CPU time: 34.03 s, Wall time: 99.62 s, Max memory usage: 675.11
MB.
902 [2026-06-11 17:53:12] [3149297:info] 10000 ++++++
903 [2026-06-11 17:53:12] [3149297:info] 10000 VCOM runtime: 0:1:39
904 [2026-06-11 17:53:12] [3149297:info] 10000 ++++++
:set nu 904,1 Bot
```

图 9: Compilation Results

5. PnR 生成比特流

进入 fpgaCompDir/ 目录，运行 make all 命令进行 PnR 生成比特流文件。

```
chenyiheng@ubuntu20-08:~/g13g14/z24301050026/lab10/wujian100_p2e$ cd fpgaCompDir/
chenyiheng@ubuntu20-08:~/g13g14/z24301050026/lab10/wujian100_p2e/fpgaCompDir$ ll
total 20
drwxr-xr-x 3 chenyiheng nas 4096 6月 11 18:52 ./
drwxr-xr-x 11 chenyiheng nas 4096 6月 11 18:19 ../
-rw-r--r-- 1 chenyiheng nas 3952 6月 11 17:53 Makefile
drwxr-xr-x 3 chenyiheng nas 4096 6月 11 18:30 part_b0_f0/
-rw-r--r-- 1 chenyiheng nas 0 6月 11 18:30 .part_b0_f0_syn.complete
-rw-r--r-- 1 chenyiheng nas 22 6月 11 17:53 user_pnr_constraints.tcl
chenyiheng@ubuntu20-08:~/g13g14/z24301050026/lab10/wujian100_p2e/fpgaCompDir$ make
rm -rf *.complete
make -f /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/fpgaCompDir/part_b0_f0/Makefile syn > /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/fpgaCompDir/part_b0_f0/.syn.log & wait
sh: 1: /bin/rpm: not found
rm -rf *.pnr.complete
/opt/xepic/hpe-24.09.01.s003/tools/python3.11.3/bin/python3 /opt/xepic/hpe-24.09.01.s003/share/pnr/scripts/farm_check_syn_status.py
make -f /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/fpgaCompDir/part_b0_f0/Makefile pnr pnr_strategy=default n_jobs=7 apply_best_result=0 mpar_cluster_config=local > /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/fpgaCompDir/part_b0_f0/.pnr.log & wait
sh: 1: /bin/rpm: not found
^R
/opt/xepic/hpe-24.09.01.s003/tools/python3.11.3/bin/python3 /opt/xepic/hpe-24.09.01.s003/share/pnr/scripts/farm_check_pnr_status.py
/opt/xepic/hpe-24.09.01.s003/tools/python3.11.3/bin/python3 /opt/xepic/hpe-24.09.01.s003/share/pnr/scripts/pnr_summary.py

*****
PNR_summary_table:
*****
+-----+-----+-----+-----+-----+-----+-----+-----+
| Partitions | Total_PNR_runtime | Initial_PNR_runtime | MPAR_runtime | PNR_result | Timing_result | WNS | TNS | WHS |
| THS | WPWS | TPWS | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+
| part_b0_f0 | 0:54:7 | 0:54:7 | - | Bitfile generated | Timing met | 0.481 | 0.000 | 0.010 |
| 0 | 0.000 | 0.000 | 0.000 | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+

/opt/xepic/hpe-24.09.01.s003/bin/postPrTiming -vcom_data_dir /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e -fpga_data_dir /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e/fpgaCompDir -div_rootClk 1000 -output_dir /home/data/chenyiheng/g13g14/z24301050026/lab10/wujian100_p2e -single_board -fast_user_clock 20
```

图 10: PnR Generation

6. vDbg 上板测试

执行 vdbg vdbg.tcl, 使用 vDbg 进行上板测试, 可以看到生成了波形文件。

```
chenyiheng@ubuntu20-08:~/g13g14/z24301050026/lab10/wujian100_p2e/fpgaCompDir$ cd ..
chenyiheng@ubuntu20-08:~/g13g14/z24301050026/lab10/wujian100_p2e$ vdbg vdbg.tcl
VDBG
v24.09.01.s003 x86_64 Dec.20 2024

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This program is proprietary and confidential information of XEPIC Co., Ltd.
and may be used and disclosed only as authorized in a license agreement
controlling such use and disclosure.

Checkout license successfully. The remaining valid days: 6383.
1/3 set clock and voltage 100%
2/3 download bin file 100%
3/3 set default configuration 100%
INFO vdbg-11043: Download successfully! Design is ready to run.
INFO vdbg-10758: Set trace size is successfully.
INFO vdbg-10735: Open waveform database wujian100_wave.vcd successfully.
INFO vdbg-15721: Create physical probe ph0.
INFO vdbg-15721: Create physical probe ph1.
INFO vdbg-15721: Create physical probe ph2.
INFO vdbg-15721: Create physical probe ph3.
INFO vdbg-15721: Create physical probe ph4.
INFO vdbg-15721: Create physical probe ph5.
INFO vdbg-15721: Create physical probe ph6.
INFO vdbg-15721: Create physical probe ph7.
INFO vdbg-15721: Create physical probe ph8.
INFO vdbg-15721: Create physical probe ph9.
INFO vdbg-15721: Create physical probe ph10.
INFO vdbg-15721: Create physical probe ph11.
INFO vdbg-15721: Create physical probe ph12.
DUT reset completed
Uploading waveform...
INFO vdbg-15540: Upload waveform successfully.
INFO vdbg-17182: Physical probe disabled
Debug finished. File: wujian100_wave.vcd
INFO vdbg-13058: Releasing hardware resource and clear fpga, wait a moment...
INFO vdbg-11245: vdbg exited at Jun 11 2026 21:06:09 exitcode: 0.
chenyiheng@ubuntu20-08:~/g13g14/z24301050026/lab10/wujian100_p2e$ ll *.vcd
-rw-r--r-- 1 chenyiheng nas 1791641 6月 11 21:06 wujian100_wave.vcd
chenyiheng@ubuntu20-08:~/g13g14/z24301050026/lab10/wujian100_p2e$
```

图 11: vDbg Test

7. 初步波形分析

将生成的波形文件下载到本地, 并使用 GTKWave 打开进行分析. 将 `monitor_hwdata[31:0]` 的 Data Format 设置为 ASCII, `monitor_haddr[31:0]` 保持默认 Hex, 对后者按照 String 搜索 `2001FFF0`. 并未观察到 C 程序中 `printf("Hello World!\n");` 的预期输出, 而是显示 `Exception: NO.4` 等异常信息. 说明程序执行发生了异常, 需要进一步分析.

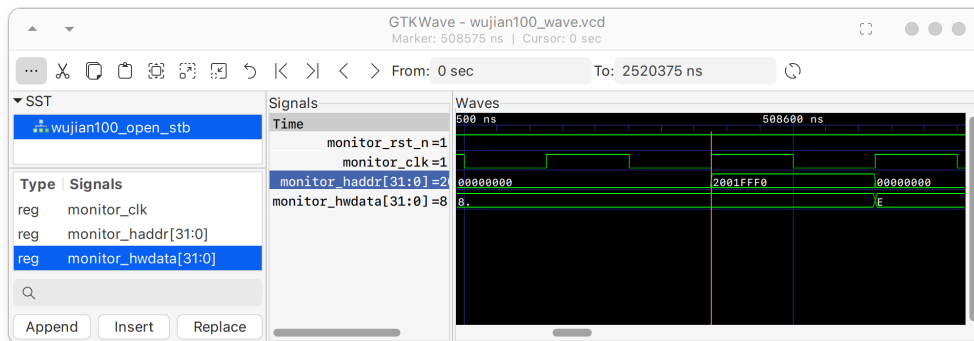


图 12: Waveform Error 01

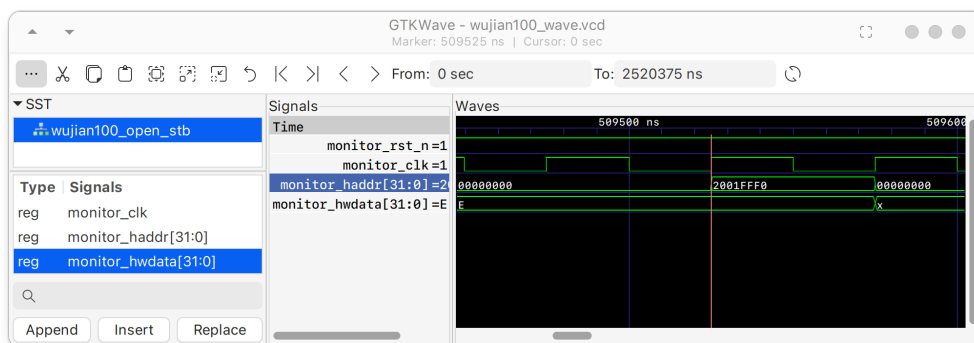


图 13: Waveform Error 02

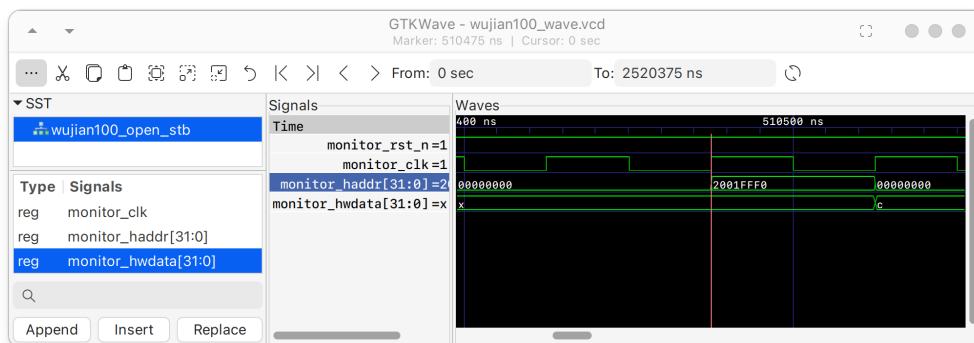


图 14: Waveform Error 03

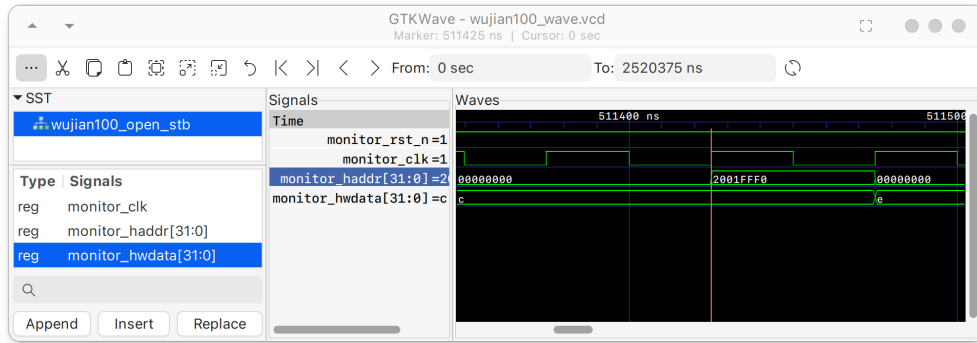


图 15: Waveform Error 04

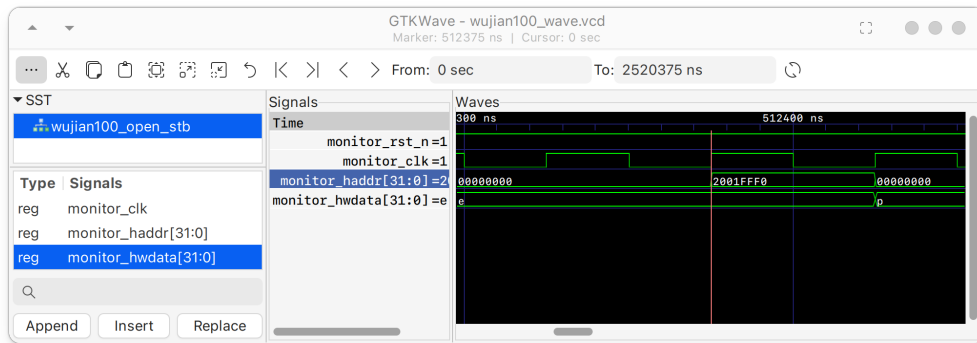


图 16: Waveform Error 05

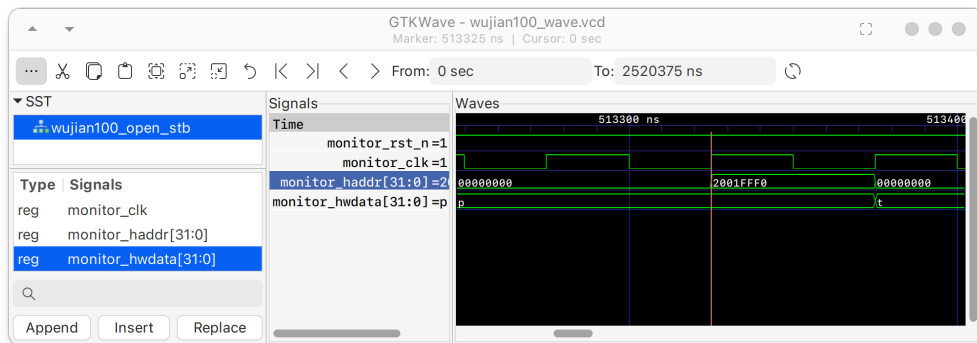


图 17: Waveform Error 06

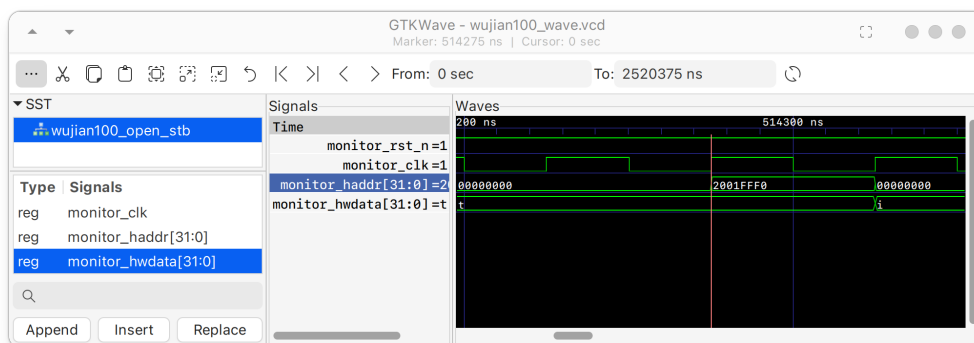


图 18: Waveform Error 07

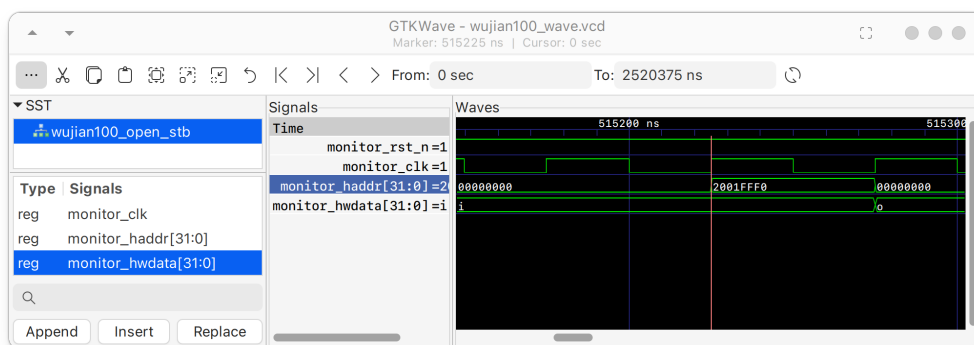


图 19: Waveform Error 08

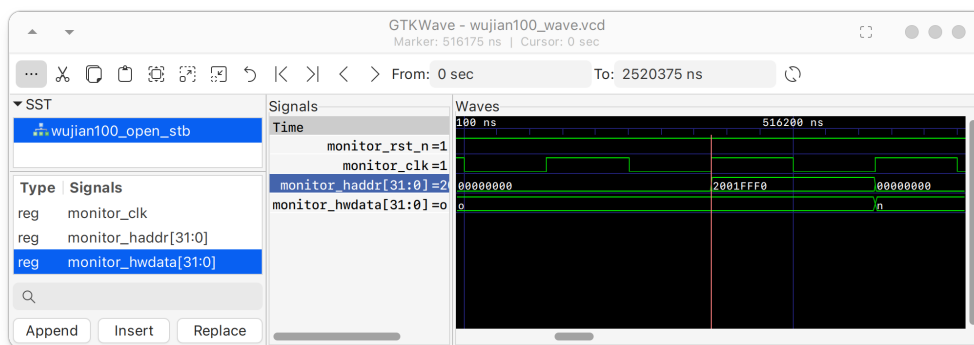


图 20: Waveform Error 09

8. 波形分析方法

为获取实际信号波形, 首先需要在 vCom 编译阶段将关键信号标记为硬件级的物理探针, 随后在 vDbg 运行期间按时钟周期实时采集信号跳变并将数据暂存至板载存储器中, 测试结束后再将缓存数据上传至主机。

采用这种方法是因为 FPGA 内部的高速信号很难直接从外部捕捉, 只能通过片内探针来获取真实物理波形, 同时按需捕捉以平衡资源消耗和观测需求。这提供了最接近真实的信号状态, 避免了纯软件仿真可能存在的假象。

9. 错误原因分析

经排查发现, 实验提供的 `elf2mem.py` 脚本的间隙填充逻辑中存在错误, 导致在 ELF 的每个地址空洞处都会丢失一个 32 位字节. 而本次实验使用的 CDK 编译生成的 `.elf` 文件恰好包含一个这样的空洞, 即 `.eh_frame` (结束于 `0x49EC`) 和 `.rodata` (起始于 `0x49F0`) 之间的一个 4 字节对齐间隙. 因此, 从 `.rodata` 开始的所有内容都被加载到了比其链接地址低 4 个字节的位置, 这导致了 CPU 在后续执行过程中触发了加载地址未对齐异常, 并没有执行到 `main()` 函数. 此前捕获的波形实际上是这次崩溃的过程, 而不是程序的正常输出.

转换脚本 `elf2mem.py` 的 `hex2mem()` 函数中, `cur_mem_addr` 会在每次写入一个字后进行后置递增, 然后保存 `prev_mem_addr = cur_mem_addr`. 因此, `prev_mem_addr` 保存的实际上是下一个需要写入的字的索引. 当遇到空洞后的下一条记录开始时, 缺失的间隙字数恰好为 `mem_addr_incr = cur_mem_addr - prev_mem_addr`, 此时需要输出全零的字的个数也就等于 `mem_addr_incr`. 然而, 原始代码认为 `prev_mem_addr` 代表最后一个写完的字, 因此对空洞采用了如下写入逻辑, 导致每次遇到空洞时都会少写入一次.

```
if mem_addr_incr > 1:
    for i in range(mem_addr_incr - 1):
        ofile_id.write("00000000\n")
```

这可以在任何现有的文件上通过 `wc -l` 命令进行验证。生成的镜像跨度为 `0x0-0x5C00`，共 5888 个字。而生成的 `hello_world.mem` 文件则只有 5887 行，少了一个字。正确的逻辑只需要修改两个符号即可。

```
if mem_addr_incr > 0:
    for i in range(mem_addr_incr):
        ofile_id.write("00000000\n")
```

修改后采用同样的 .elf 文件重新生成, wc -l 的结果变为了正确结果 5888。通过 vDbg 重新加载这分割后的 4 个 .mem 文件后上板验证, 异常写入操作消失, 0x2001FFF0 地址成功接收到 13 次写入操作, 即 main 函数中输出的 Hello World!\n。

该 bug 具有很强的隐蔽性, 仅在加载镜像包含地址空洞的 .elf 文件上才会触发. 不幸的是, 该程序在 CDK 中生成的布局总是会产生一个空洞, 因此本实验的每一次作业提交都受到了影响. 并且这种故障模式极具欺骗性, 虽然产生了看似合理的波形输出, 但实际上得到的只是一行崩溃信息. 建议及时修改服务器上提供的 elf2mem.py 脚本文件, 即使这并不影响整体的实验流程.

10. 最终波形分析

修改脚本后重新生成 .mem 文件并上板测试, 可以看到 0x2001FFF0 地址成功接收到了 Hello World!\n. 这说明程序已经正确执行到了 main() 函数, 并且成功调用了 printf() 输出了预期的字符串.

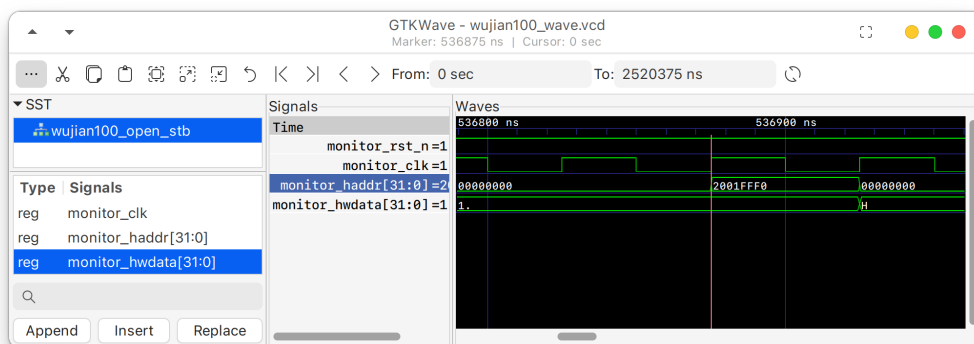


图 21: Waveform Success 01

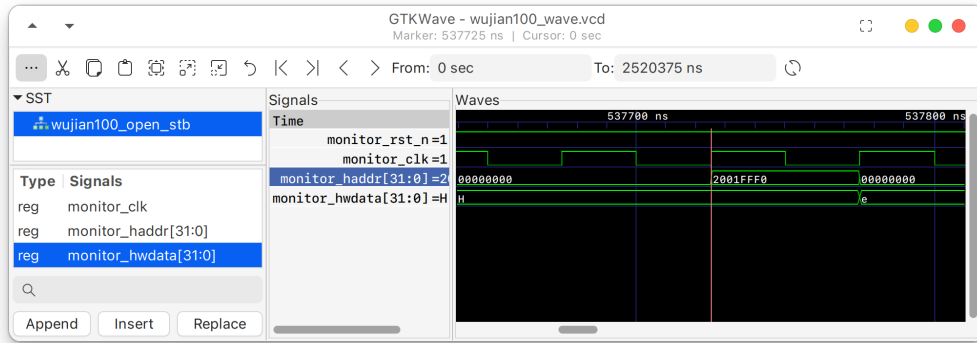


图 22: Waveform Success 02

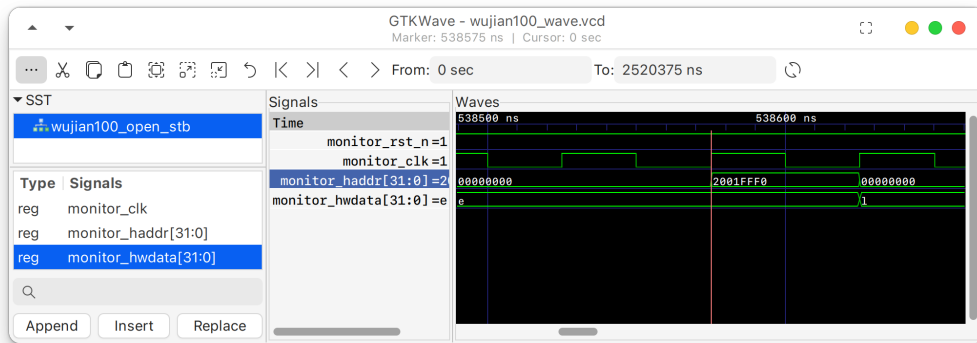


图 23: Waveform Success 03

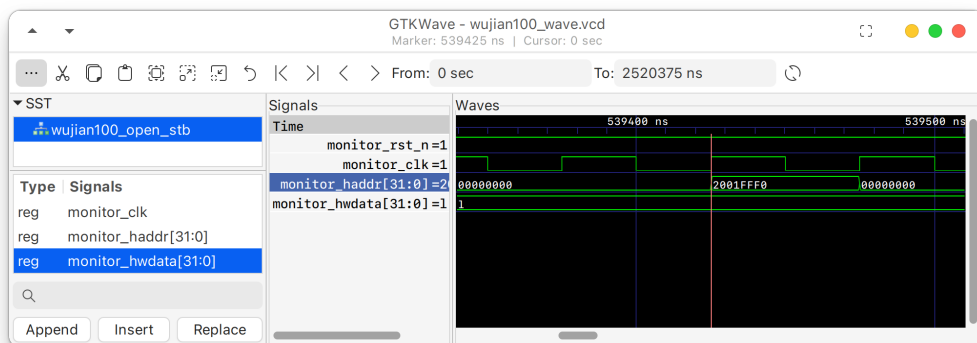


图 24: Waveform Success 04

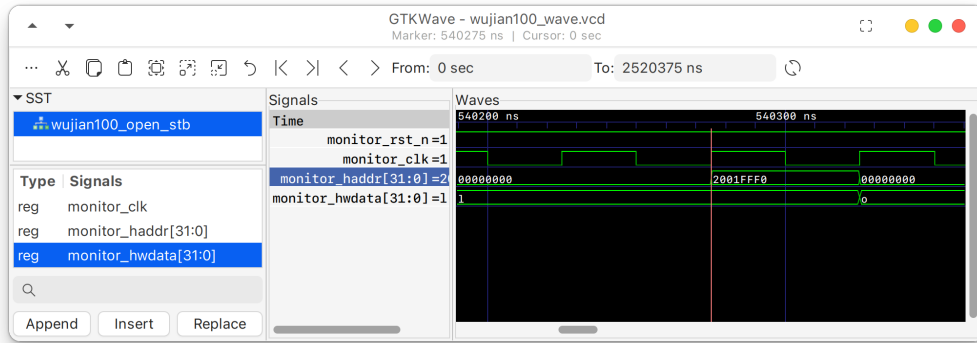


图 25: Waveform Success 05

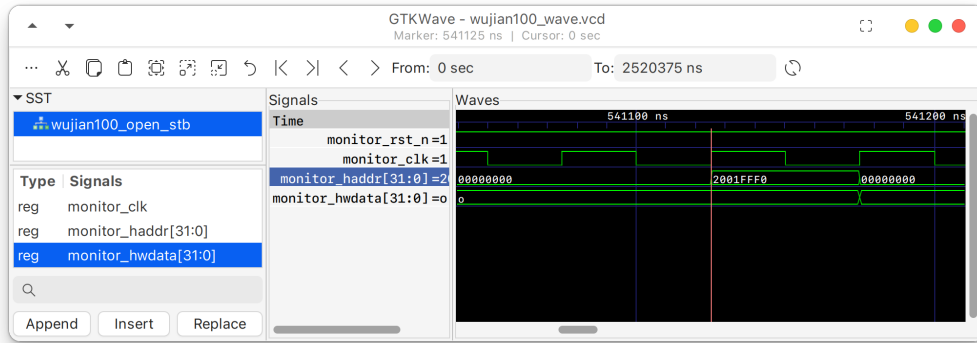


图 26: Waveform Success 06

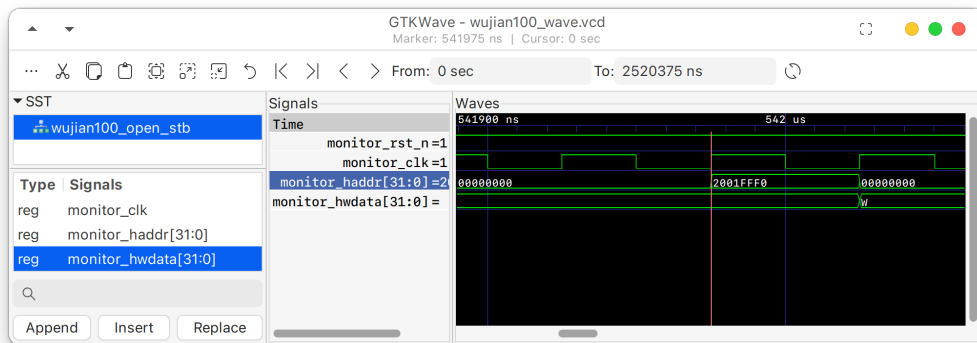


图 27: Waveform Success 07

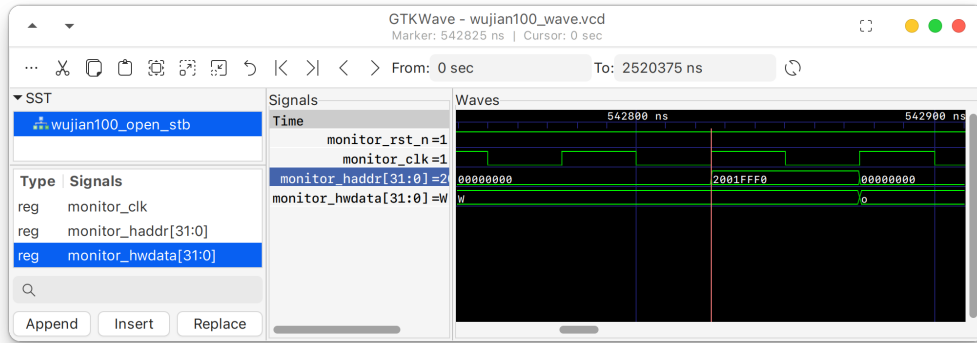


图 28: Waveform Success 08

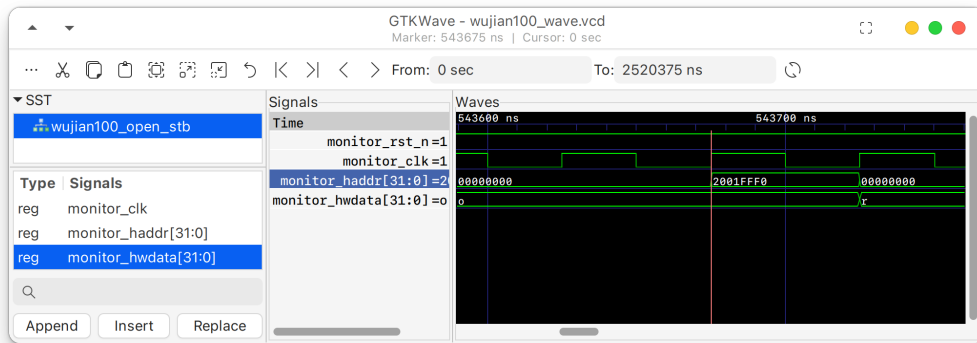


图 29: Waveform Success 09

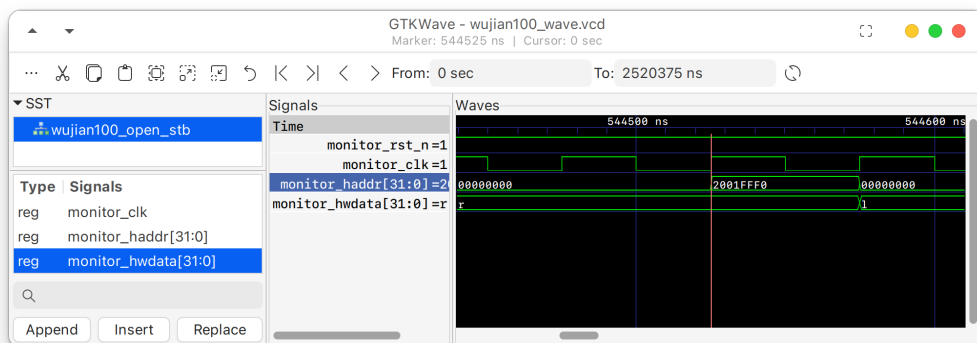


图 30: Waveform Success 10

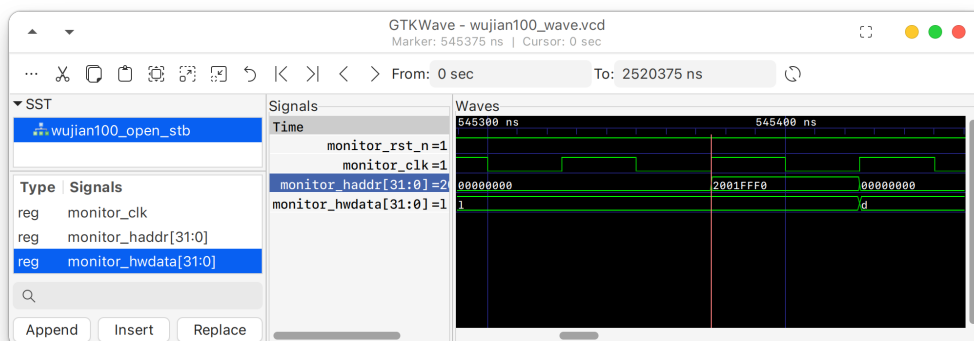


图 31: Waveform Success 11

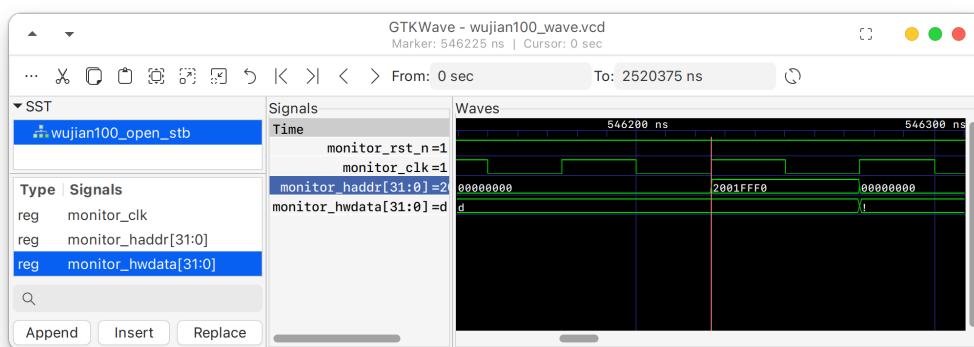


图 32: Waveform Success 12

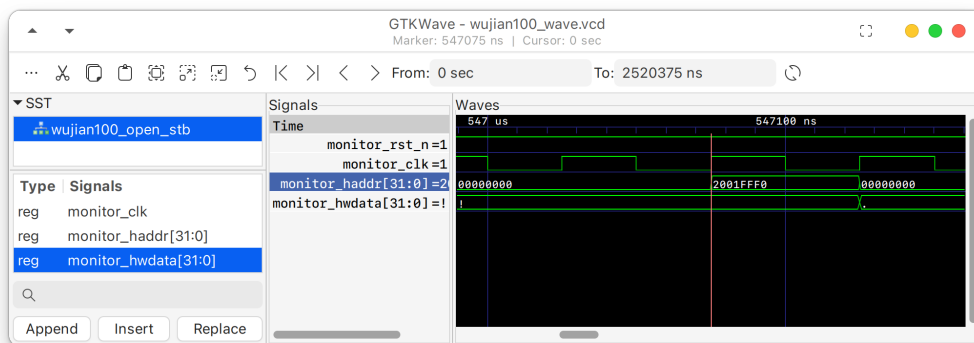


图 33: Waveform Success 13

2.3. 自行编写一个 C 语言的简单算法程序, 进行仿真测试, 步骤参考“三、wujian100 SoC 的 FPGA 原型测试-5”, 对测试结果进行截图, 并说明其正确性。

1. 项目编译

编写一个简单的快速排序 C 程序, 进行仿真测试. 程序中定义了一个整数数组, 并调用快速排序函数对其进行排序. 最后通过 printf() 输出排序后的结果. 使用 CDK 编译生成 .elf 文件。

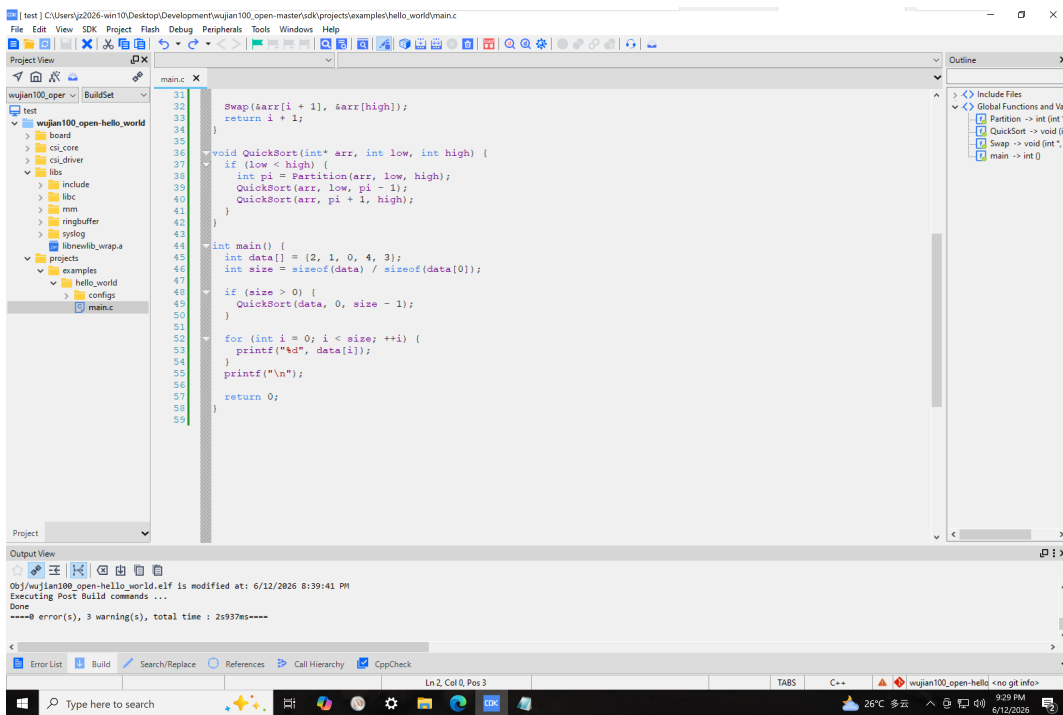


图 34: Compile ELF

2. 测试结果

将生成的 .elf 文件上传到服务器, 保持文件命名不变, 使用更新后的 Python 脚本转换并分割. 由于之前已经执行过综合, 编译并生成比特流, 因此可以直接使用 vDbg 进行上板测试. 测试成功后观察生成的波形文件, 可以看到程序成功输出了排序后的数组, 说明程序执行正确.

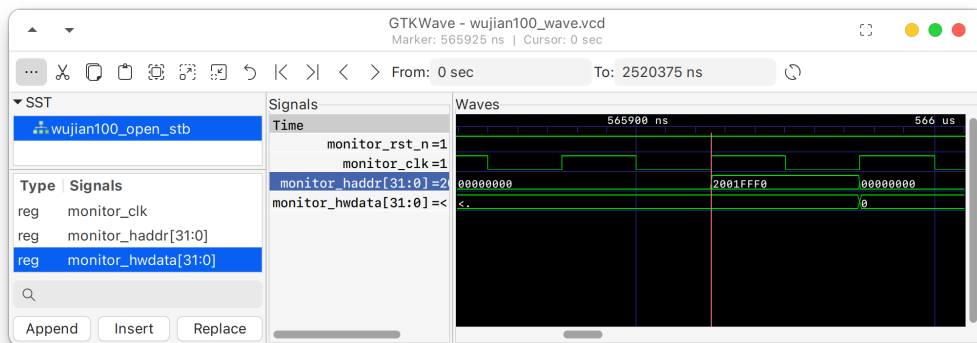


图 35: Quick Sort Results

3. 实验分析与总结

本次实验主要围绕基于 MCU 的 wujian100 SoC 平台展开, 深入剖析了从软件 C 代码编译到 FPGA 硬件原型的完整验证工作流. 在此过程中不仅理解了 CPU, 多级总线与各类外设的系统互联架构, 更掌握了 EDA 工具的底层运行逻辑与硬件调试技巧. 通过动手排查高度隐蔽的地址错位 Bug, 极大提升了阅读底层指令, 理解内存映射机制以及使用 GTKWave 观察波形的实战分析能力. 这种面对由上层软件引发底层硬件执行崩溃的跨域调试经验, 打破了纯软件或者纯硬件视角的局限, 得以更加全面地理解 SoC 设计与验证的复杂性, 以及软硬件协同调试在现代计算系统开发中的重要性.

总体而言,本实验完整地进行了 wujian100 SoC 的前端仿真与基于芯华章 P2E 工具链的 FPGA 原型测试流程. 实验依次完成了基于 CDK 的 C 语言交叉编译, 物理内存镜像文件的转换与分割, vSyn 综合, vCom 编译, 以及 PnR 比特流生成, 最终使用 vDbg 工具完成了真实的硬件上板调试. 在成功定位并修复了内存转换工具的 bug 后, 实验不仅顺利完成了标准字符输出的测试, 还实现了快速排序算法的硬件级原型验证, 最终通过正确的逻辑波形输出了排序结果, 使得整个软硬件协同设计的闭环得到了严格且全面的检验.

4. 实验收获与建议

此次实验不仅提供了从顶层软件编译到底层硬件验证的完整流程, 更加深了对处理器架构与 EDA 工具链的理解, 也大幅提升了应对软硬件接口的跨层级调试能力. 通过动手操作全过程中的核心环节, 不仅掌握了前端仿真与物理硬件测试的方法差异, 更深入探索了底层内存空间布局, 目标文件段对齐, 以及 CPU 硬件异常陷阱的触发机制等问题. 最深刻的体会在于, 系统级开发绝不能盲从既有的中间工具链, 当硬件执行状态与预期产生分歧时, 必须具备深入源码排查的怀疑精神, 以确保指令执行的绝对可靠.

针对实验中暴露出的工具链问题, 建议在实验平台的服务器环境中尽早修复 elf2mem.py 的地址空洞填充逻辑. 此外, 鉴于纯粹依靠肉眼在 GTKWave 中排查指令集级别的异常难度较高且极耗费时间, 而直接处理 .vcd 波形文本文件又需要对其格式和内容有较为深入的理解, 建议在实验文档中增加相关指南或提供相关工具, 以帮助学生专注于对有效信息的分析, 提升实验体验.